

The Carbon Family Contd...Important Compounds

1. Carbon monoxide, CO . used as a **gaseous fuel**, as a **precursor for organic compounds**, and as a **reactant in the purification of nickel**. Formed in internal combustion engines and released as a toxic air pollutant. *When something burns, if there's no enough sufficient oxygen there CO may produced and it is too toxic.*
2. Carbon dioxide, CO_2 . Atmospheric component used by photosynthetic plants to make carbohydrates and O_2 . The final oxidation product of all carbon-based fuels; its increase in the atmosphere is changing Earth's climate. Used industrially as a refrigerant gas, a blanketing gas in fire extinguishers, and an effervescent gas in beverages. Combined with NH_3 to form urea for fertilizers and plastics manufacture.
3. Methane, CH_4 . Used as a fuel and in the production of many organic compounds. Major component of natural gas. Formed by anaerobic decomposition of plants (swamp gas) and by microbes in termites and certain mammals. Greenhouse gas that contributes to global warming. *80% of natural gases*

One problem of using CH_4 as a fuel is that it is dangerous to transport CH_4 as it is highly flammable. Though, we can convert CH_4 into methanol and transport as a liquid.

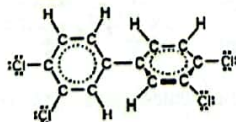
Highlights of Carbon Chemistry

- Carbon forms bonds with the smaller Group 1A(1) and 2A(2) metals, many transition metals, the halogens, and its neighbors B, Si, N, O, P, and S.
- It also exhibits every oxidation state possible for its group, from +4 in CO_2 and halides like CCl_4 through -4 in CH_4 .
- Two features stand out in the chemistry of carbon: ability to bond to itself to form chains and to form multiple bonds.
- As a result of its small size and its capacity for four bonds, carbon can form chains, branches, and rings that lead to myriad structures. Add a lot of H, some O and N, a bit of S, P, halogens, and a few metals, and you have the whole organic world!

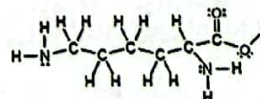
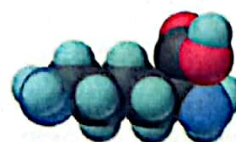
these compounds are known as Organo-metallic compounds



Acrylonitrile



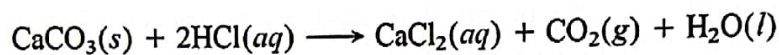
PCB



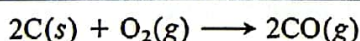
Lysine

Highlights of Carbon Chemistry

- Multiple bonds are common in carbon structures because the C-C bond is short enough for side-to-side overlap of two half-filled 2p orbitals to form π bonds.
- Because the other 4A members are larger, E-E bonds become longer and weaker down the group; thus, in terms of bond strength, $C-C > Si-Si > Ge-Ge$. The empty d orbitals of these larger atoms make the chains much more susceptible to chemical attack.
- Thus, some compounds with long Si chains are known but they are very reactive, and the longest Ge chain has only eight atoms.
- Moreover, longer bonds do not typically allow sufficient overlap of p orbitals for π bonding. Only very recently have compounds with π bonds between Si atoms or between Ge atoms been prepared, and they are also extremely reactive.
- In contrast to its organic compounds, carbon's inorganic compounds are simple. Metal carbonates are the main mineral form. Marble, limestone, chalk, coral, and several other types are found in enormous deposits throughout the world.
- Many of these compounds are leftovers of fossilized marine organisms. Carbonates are used in several common antacids because they react with the HCl in stomach acid



- Unlike the other 4A members, which form only solid network-covalent or ionic oxides, carbon forms two common gaseous oxides, CO_2 and CO .
- Carbon dioxide is essential to all life: it is the primary source of carbon in plants and animals through photosynthesis.
- Its aqueous solution is the cause of acidity in natural waters.
- However, its atmospheric buildup from deforestation and excessive use of fossil fuels is severely affecting the global climate.
- Carbon monoxide forms when carbon or its compounds burn in an inadequate supply of O_2

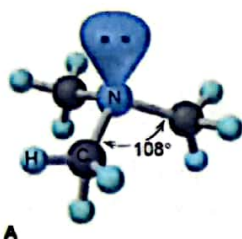


- Carbon monoxide is a key component of syngas fuels and is widely used in the production of methanol, formaldehyde, and other major industrial compounds.
- CO binds strongly to many transition metals. When inhaled in cigarette smoke or polluted air, it enters the blood and binds strongly to the $Fe(II)$ in hemoglobin, preventing the normal binding of O_2 , and to other iron-containing proteins.
- The cyanide ion (CN^-) is isoelectronic with CO :



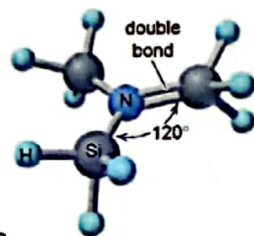
Highlights of Silicon Chemistry

- Silicon halides are more reactive than carbon halides because Si has empty d orbitals that are available for bond formation.
- One explanation for this unusual strength is that the Si-X bond has some double-bond character because of the presence of a σ bond and a different type of π bond, called a $p, d-\pi$ bond. This bond arises from side-to-side overlap of an Si d orbital and a halogen p orbital.
- A similar type of $p, d-\pi$ bonding leads to the trigonal planar molecular shape of trisilylamine, in contrast to the trigonal pyramidal shape of trimethylamine

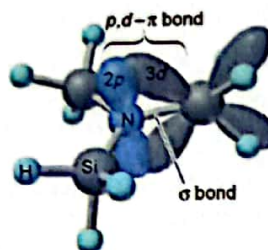


A

methy lamonia



B



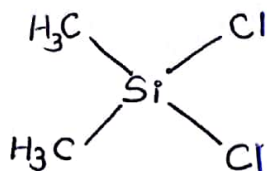
C

- Just as carbon forms unending C-C chains, the Si-O grouping repeats itself endlessly in a wide variety of **silicates**, the most important minerals on the planet, and in **silicones**, synthetic polymers that have many applications:

1. Silicate minerals.

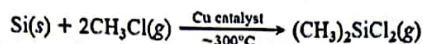
- Oxygen, the most abundant element on Earth, and silicon, the next most abundant, compose these minerals and thus account for four of every five atoms on the surface of the planet!



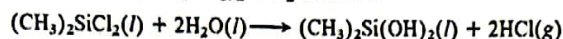


2. Silicone polymers

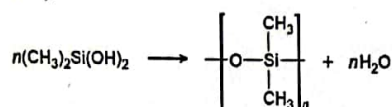
- A key starting material is formed by the reaction of silicon with methyl chloride:



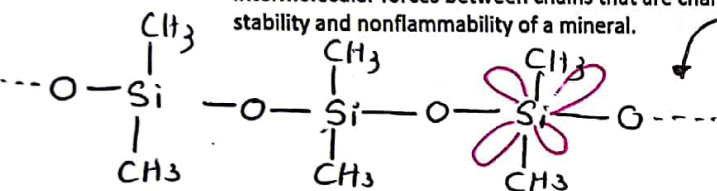
Although the corresponding carbon compound $[(\text{CH}_3)_2\text{CCl}_2]$ is inert in water, the empty d orbitals of Si make $(\text{CH}_3)_2\text{SiCl}_2$ reactive:



The product reacts with itself to form water and a silicone chain molecule called a *poly(dimethyl siloxane)*:



- Silicones have properties of both plastics and minerals. The organic groups give them the flexibility and weak intermolecular forces between chains that are characteristic of a plastic, while the O-Si-O backbone confers the thermal stability and nonflammability of a mineral.



Because of the presence of CH_3- (methyl) groups this compound have organic properties. Also the $\text{Si}-\text{O}$

backbone gives this compound mineralic properties as well.

Because of this, we can ~~try~~ heat for a comparatively higher temperature without damaging the structure

Diagonal Relationships: Boron and Silicon

metalloids boron and silicon.

- Both of these elements exhibit the electrical properties of a semiconductor
- Both B and Si and their mineral oxoanions (i.e; borates and silicates) occur in extended covalent networks.
- Both boric acid $[\text{B(OH)}_3]$ and silicic acid $[\text{Si(OH)}_4]$ are weakly acidic solids that held together by H bonding.
- Their hydrides (boranes silanes) are both flammable, low-melting compounds that act as reducing agents.

Why something has a stretchy property

The Halogens

Group 7A(17): The Halogens

KEY

Atomic No.
Symbol
Atomic mass
Valence e⁻
Configuration
Common oxidation states

GROUP 7A(17)

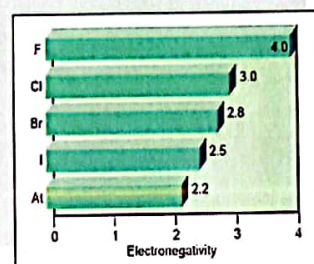
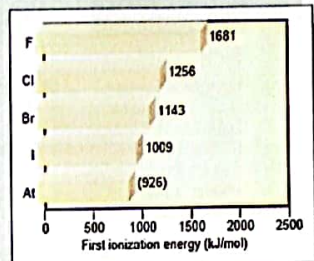
ns^2np^5

Atomic Properties

Atomic radius (pm)	Ionic radius (pm)
F 72	F ⁻ 133
Cl 100	Cl ⁻ 181
Br 114	Br ⁻ 196
I 133	I ⁻ 220
At (140)	no data

Group electron configuration is ns^2np^5 ; elements lack one electron to complete their outer level. The -1 oxidation state is the most common for all members. Except for F, the halogens exhibit all odd-numbered states (+7 through -1).

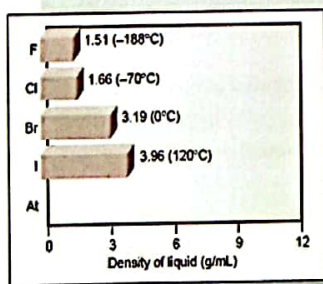
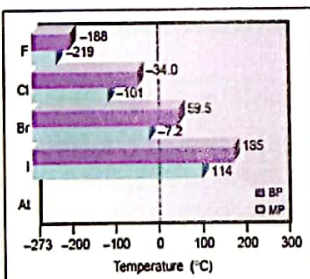
Down the group, atomic and ionic size increase steadily, as IE and EN decrease.



9 F 19.00 $2s^22p^5$ (-1)	Photograph not available
17 Cl 35.45 $3s^23p^5$ (-1, +7, +5, +3, +1)	
35 Br 79.90 $4s^24p^5$ (-1, +7, +5, +3, +1)	
53 I 126.9 $5s^25p^5$ (-1, +7, +5, +3, +1)	
85 At (210) $6s^26p^5$ (-1)	Extremely rare, no sample available

Physical Properties

Down the group, melting and boiling points increase smoothly as a result of stronger dispersion forces between heavier molecules.

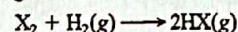


The densities of the elements as liquids (at given T) increase steadily with molar mass.

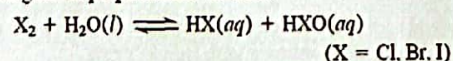
Important Reactions

Oxidizing strength and aqueous redox chemistry of the halogens are shown in reactions 1 and 2, and industrial processes involving fluorine in reactions 3 and 4.

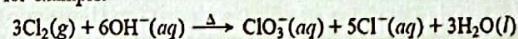
1. The halogens (X_2) oxidize many metals and nonmetals. The reaction with hydrogen, although not used commercially for HX production (except for high-purity HCl), is characteristic of these strong oxidizing agents:



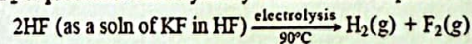
2. The halogens disproportionate in water:



In aqueous base, the reaction goes to completion to form hypohalites (see text) and, at higher temperatures, halates; for example:

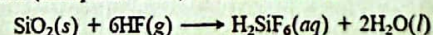


3. F_2 is produced electrolytically at moderate temperature:



A major use of F_2 is in the preparation of UF_6 for nuclear fuel.

4. Glass (amorphous silica) is etched with HF:



Important Compounds

1. Fluorspar (fluorite), CaF_2 . Widely distributed mineral used as a flux in steelmaking and in the production of HF (see photo).

2. Hydrogen fluoride, HF. Colorless, extremely toxic gas used to make F_2 , organic fluorine compounds, and polymers. Also used in aluminum manufacture and in glass etching (margin note, p. 614).

3. Hydrogen chloride, HCl. Extremely water-soluble gas that forms hydrochloric acid, which occurs naturally in the stomach fluids of mammals (humans produce 1.5 L of 0.1 M HCl daily) and in volcanic gases (from reaction of H_2O and sea salt). Made by reaction of NaCl and H_2SO_4 and as a byproduct of plastics (PVC) production. Used in the "pickling" of steel (removal of adhering oxides) and in the production of syrups, rayon, and plastic.

4. Sodium hypochlorite, NaClO , and calcium hypochlorite, $\text{Ca}(\text{ClO})_2$. Oxidizing agents used to bleach wood pulp and textiles and to disinfect swimming pools, foods, and sewage (also used to disinfect the Apollo 11 on return from the Moon). Household bleach is 5.25% NaClO by mass in water.

5. Ammonium perchlorate, NH_4ClO_4 . Strong oxidizing agent used in the space shuttle program (see photo).

6. Potassium iodide, KI. Most common soluble iodide. Added to table salt to prevent thyroid disease (goiter). Used in chemical analysis because it is easily oxidized to I_2 , which provides a colored end point.

7. Polychlorinated biphenyls, PCBs. Mixture of chlorinated organic compounds used as nonflammable insulating liquids in electrical transformers. Production discontinued because of persistence in the environment, where they become concentrated in the tissues of fish, birds, and mammals, and cause reproductive disturbances and possibly cancer.



Fluorite



Space shuttle

I. Electronic Configurations.

F ($Z = 9$) - ($2s^2p^5$)

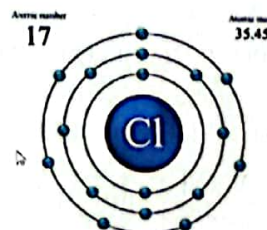
Cl ($Z = 17$) - ($3s^2p^5$)

Br ($Z = 35$) - ($4s^2p^5$)

I ($Z = 53$) - ($5s^2p^5$)

At ($Z = 85$) - ($6s^2p^5$)

- All the halogens have seven electrons (ns^2p^5) in their outer most shell.
- Two electrons are in ns -orbital and the remaining five electrons are in np -orbitals and are distributed as np^2_x , np^2_y , np^1_z in accordance with Hund's rule.
- The shell previous to the outermost shell contains in F the group s^2 (saturated), in Cl s^2p^6 (saturated), in Br $s^2p^6d^{10}$ (saturated) while in I and At $s^2p^6d^{10}$ (unsaturated).
- This shows why F differs from Cl and these two from the remaining elements. Since the additional electron in these elements enters the p orbital of the outermost shell, these are called p -block elements of the long form of the periodic table.



Similarities in Physical Properties

1. Occurrence.

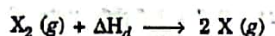
These elements do not occur in the free state. But in the combined state they occur fairly abundantly in nature.

2. Diatomic Nature.

- Exist as covalent diatomic molecules (X_2 molecule) in the gaseous, liquid and crystalline states.
- These X_2 molecules are held together by weak Van der Waals forces which explain the volatile nature of these elements.
- At high temperatures, X_2 molecules dissociate into X atoms.
- The stability of diatomic molecules towards heat decreases from F_2 to I_2 .

3. Heat of Dissociation of X_2 Molecules.

- Heat of dissociation (H_d) of X_2 molecule is the amount of heat required to dissociate X_2 molecule in the gaseous state into X gaseous atoms or to break X—X bond in X_2 molecule.



Obviously, ΔH_d is equal to X—X bond energy. Excepting for F_2 these energies decrease from Cl_2 to I_2 .

-The lower value of heat of dissociation of F_2 is due to the reasons;

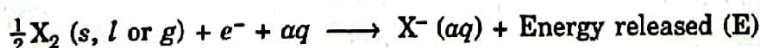
(i) Due to very small F—F bond length, repulsion between the non-bonding electrons is very large as compared to other X—X bond lengths (X = Cl, Br, I). This makes the F atoms in F_2 molecule repel each other and helps the dissociation of F_2 molecule into F atoms.

(ii) X—X bond in Cl_2 , Br_2 and I_2 molecules is stronger than F—F bond in F_2 molecule. This is due to the possibility of the existence of multiple bonding in X—X bond involving d-orbitals.

(Also this can oxidize other other halogens)

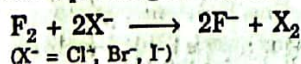
F^- is the most stable anion. so it should be the

4. Oxidizing Property. A substance that has a tendency to accept one or more electrons is said to show oxidizing property. The halogens, due to high electron affinity values, have a great tendency to accept an electron and hence act as strong oxidizing agent. **strongest oxidization agent.**

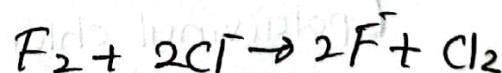
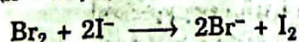
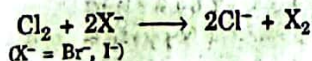


- Since the values of E are decreasing from F_2 to I_2 the oxidizing power of halogens is also decreasing in the same direction, i.e. the oxidizing power of halogens is in the order: F_2 (strongest oxidizing agent) $>$ $Cl_2 > Br_2 > I_2$ (weakest oxidizing agent).

- Since F_2 is the strongest oxidizing agent in the series, it will oxidize other halide ions to halogens in solution or when dry, i.e. F_2 displaces other halogens from their corresponding halides.



- Similarly Cl_2 will displace Br^- and I^- ions from their solutions and Br_2 will displace I^- ions from their solutions.



5. Formation of Clathrates.

Water solutions of the halogens on cooling give the crystals of their clathrates of $X_2 \cdot xH_2O$ type. Examples are : $Cl_2 \cdot 7.3 H_2O$, $Br_2 \cdot 10H_2O$, $I_2 \cdot xH_2O$ etc.

6. Electronegativity.

Halogens have large values of electronegativity. These values decrease from F to I in the group. Large electronegativity values of halogen atoms indicate that X atoms have a strong tendency to form X^- ions.

7. Solubility.

- The halogens are slightly soluble in water and their solubility decreases from Cl to I.
- F cannot be dissolved in H_2O , since it decomposes H_2O liberating O_2 and some O_3 .
- The halogens dissolve much better than H_2O in many organic solvents like carbon disulphide, chloroform, ether etc.
- In organic solvents, iodine gives violet colour, bromine yields brown colour and chlorine produces yellow colour.
- It is assumed that in organic solvents halogens exist as free molecule as in the gas phase. In nucleophilic (electron donating) polar solvent like alcohols, ketones the halogens give coloured solutions whose colours are due to the formation of solvent – halogen type complexes. These complexes are charge-transfer compounds.

2nd Slide

4. Silicon dioxide, SiO_2 : Occurs in many amorphous (glassy) and crystalline forms, quartz being the most common.

Used to

make glass and as an inert chromatography support material.

5. Silicon Carbide, SiC : Known as Carborundum, a major industrial abrasive and a highly refractory ceramic for tough, high temperature uses. Can be doped to form a high-temperature semiconductor.

6. Organotin compounds, R_4Sn : Used to stabilize PVC [poly(vinyl chloride)] plastics and to cure Silicone rubbers.

7. Tetraethyl Lead $(C_2H_5)_4Pb$: Once used as a gasoline additive to improve fuel efficiency, but now banned because of its inactivation of auto catalytic converters. Major source of Lead as a toxic air pollutant.